JAMAL MAMMADZADA

STUDENT RESEARCHER / MACHINE LEARNING SYSTEMS / SECURITY

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RESEARCH PROFILE

High school researcher and engineer building auditable systems from noisy, real-world data. Develops computer-vision pipelines, security automation, and full-stack analytical tools with human review, confidence-aware outputs, and reproducible evaluation. Interested in knowledge extraction, visual analytics, and reliable AI research infrastructure.

SELECTED TECHNICAL WORK

FRCMOB - Sole Developer **2024-Present**

- Built a full-stack FIRST Robotics analysis platform that detects and tracks robots in match video, maps camera observations into field coordinates, and converts footage into reviewable strategic evidence.
- Designed a human-in-the-loop pipeline: model outputs retain confidence, and scouts approve, correct, or reject results before they become accepted data; stack includes Python, FastAPI, React, PostgreSQL, Redis, Docker, YOLO, ByteTrack, OpenCV, OCR, and ONNX Runtime Web.

Windows Hardening Framework - Sole Developer **2025-Present**

- Engineered an approximately 11,900-line PowerShell framework for Windows and Windows Server auditing, hardening, compliance validation, and remediation across authentication, privileges, services, Defender, firewalls, and audit policy.
- Made changes evidence-first and reversible: simulate mode reports findings before modification, while backups and generated revert scripts protect destructive operations.

RESEARCH

Analytical Solvability of One-Dimensional SDEs - IB Mathematics Extended Essay **2026**

- Compared analytical solution methods across geometric Brownian motion, Ornstein-Uhlenbeck, Cox-Ingersoll-Ross, and stochastic logistic models, varying one structural feature at a time.

Temperature Dependence of Capacitor Discharge - IB Physics Investigation **2026**

- Measured RC discharge from 10-50 °C and wrote the Python analysis with gradient-bounded uncertainty; observed a 21% decrease in the time constant and tested it against the predicted Arrhenius trend.

LEADERSHIP & TECHNICAL EDUCATION

Eaglets: Robot Kits for Kids - President; Former Camp Coordinator **2024-Present**

- Lead a student-run robotics nonprofit designing custom kits, beginner-facing Python/C++ libraries, and free applied-robotics camps for grades 3–8; oversee kit readiness, recruiting, grants, partnerships, and camp operations.

Technical Instructor - iCode, United Stream Foundation, Udemy **2024-Present**

- Teach programming, machine learning, Java/OOP, and applied cryptography; authored a 35-lecture course with 16 exercises and a deliberately vulnerable capstone evaluated through hidden and mutation-based tests.

EDUCATION & DISTINCTIONS

Allen High School - International Baccalaureate Diploma Candidate **Expected June 2027**

Coursework: IB Mathematics HL II, AP Physics C: Electricity & Magnetism; prior AP Calculus BC and IB Physics SL

Honors: CyberPatriot XVIII Open Division National Semifinalist; picoCTF 2026 #68 worldwide with a perfect team score; member of FRC Team 5417 when it won the 2025 FIRST Impact Award at the Fort Worth District Event and the 2025 Tomball District Event

TECHNICAL TOOLKIT

Languages: Python, Java, C++, C, C#, JavaScript, TypeScript, PHP, PowerShell
ML & vision: YOLO, ByteTrack, OpenCV, OCR, optical flow, feature engineering, calibration, shadow deployment
Systems & data: FastAPI, React, PostgreSQL, Redis, MongoDB, Docker, Git, Windows Server, Raspberry Pi Pico, Arduino
Mathematics: Stochastic calculus, SDEs, numerical simulation, probabilistic modeling